FACE RECOGNITION: AN SVD-BASED APPROACH FOR IDENTIFYING CRIMINALS

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ABSTRACT. This project explores the application of Singular Value Decomposition (SVD) and Principal Component Analysis (PCA) in facial recognition, with a particular focus on identifying criminals.
In the first part, I use PCA to reduce the dimensionality of crime rate data from 50 U.S. states to develop a multivariate linear regression model capable of predicting the homicide rate in new states.
In the second part, I implement a facial recognition algorithm that projects images of new faces into the *face subspace*, enabling the user to identify individuals already present in a criminal database.
Keywords. Face Recognition, PCA, SVD, Linear Regression

CONTENTS

1. Part I

1.1. The dataset. The dataset (7) used in this study is represented by a matrix $\mathbb{R}^{50x7}$ with $n = 50$ U.S. states where crime rates (number of crimes per $100,000$ inhabitants) have been calculated. The variable $CRIMINE$ is *p-variate* with *p=7*:

- T_{omic}: homicide rates
- T_{rapim}: kidnapping rates
- T_{rapine}: robbery rates
- $T_{aggress}$: assault rates
- $T_{fcscasso}$: burglary rates
- T_{furti}: theft rates
- T_{fauto}: motor vehicle theft rates

We report below only the first 5 observations of the dataset (7) to illustrate its structure:

	T_{omic}	T_{rapim}	T_{rapine}	$T_{aggress}$	$T_{fcscasso}$	T_{furti}	T_{fauto}
alabama	14.2	25.2	96.8	278.3	1135.5	1881.9	280.7
alaska	10.8	51.6	96.8	284	1331.7	3369.8	753.3
arizona	9.5	34.2	138.2	312.3	2346.1	4467.4	439.5
arkansas	8.8	27.6	83.2	203.4	972.6	1862.1	183.4
california	11.5	49.4	287	358	2139.4	3499.8	663.5

TABLE 1. Reduced data matrix: 50 states and 7 quantitative variables

1.2. PCA: What is it and how does it relate to SVD.

1.2.1. *Introduction.* The p original variables are numerous, making it difficult to discern existing structures. The goal of PCA is to replace the original variables with a smaller number of artificial variables that provide a summary with minimal loss of information. Therefore, I aim to represent the observations in a space $\mathbb{R}^k \subset \mathbb{R}^p$. PCA is thus a dimensionality reduction method with low information loss.

In essence, the goal is to identify a new set of p variables $y_1, \ldots, y_p$ that are not directly observed but defined as linear combinations of the original variables $x_1, \ldots, x_p$ (which are correlated with each other), such that they explain the variance/covariance structure of $x_1, \ldots, x_p$. This involves constructing artificial variables as linear combinations of the original variables to maximize the variance of each and ensure they are **not** correlated with one another.

Therefore, I construct the artificial variables as:

$$\begin{gathered} Y_1 = a_{10} + a_{11}X_1 + \cdots + a_{1j}X_j + \cdots + a_{1p}X_p \\ \vdots \\ Y_k = a_{k0} + a_{k1}X_1 + \cdots + a_{jj}X_j + \cdots + a_{jp}X_p \\ \vdots \\ Y_p = a_{p0} + a_{p1}X_1 + \cdots + a_{pj}X_j + \cdots + a_{pp}X_p \end{gathered}$$

such that

- $Cov(Y_i, Y_k) = 0, \quad \overline{x} = \overline{y} = 0, \forall i \neq k \implies$ the artificial variables are orthogonal. In fact, in general, $Cov(x, y) = \sum_{i=1}^{n} x_i \cdot y_i = x^T y = \langle x, y \rangle$, so requiring that the correlation between two variables is zero is equivalent to requiring that their inner product is zero, which is precisely the definition of orthogonal vectors.

- $V(Y_1) \geq V(Y_2) \geq \ldots \geq V(Y_p) \implies$ the artificial variables are listed in decreasing order of variance.
- $\sum_{i=1}^{p} V(X_i) = \sum_{i=1}^{p} V(Y_i) \implies$ the total variance remains unchanged.

If we are satisfied with explaining only the majority of the variance and covariance structure, some principal components can be neglected: we consider $k^* < p$. Thus, there is a process of dimensionality reduction that does not lead to a loss of statistically relevant information.

1.2.2. *Construction of artificial variables.* To ensure that the results are not influenced by the unit of measurement and/or heteroscedasticity and/or the average order of magnitude of the observed variables, it is necessary to adopt the standardization of the original variables, as shown in (8).
Therefore, I initially transform the available observations into standardized and centered ones:

$$Z_{ij} = \frac{X_{ij} - \tilde{X}_j}{\sqrt{Var(X_j)}}$$

with $\tilde{X}_j = \frac{1}{n}\sum_{i=1}^{n} X_{ij}$, so that $E(\text{column}) = 0$ and $Var(\text{column}) = 1$.

	T_{omic}	T_{rapim}	T_{rapine}	$T_{aggress}$	$T_{fcscasso}$	T_{furti}	T_{fauto}
alabama	1.7472	-0.04963	-0.30891	0.66831	-0.36165	-1.0874	-0.50067
alaska	0.86791	2.404	-0.30891	0.72516	0.092047	0.96226	1.943
arizona	0.53171	0.78683	0.15969	1.0075	2.4377	2.4743	0.32045
arkansas	0.35068	0.17343	-0.46285	-0.078801	-0.73834	-1.1147	-1.0038
california	1.0489	2.1995	1.8439	1.4633	1.9598	1.1413	1.4787

TABLE 2. Reduced standardized data matrix: 50 states and 7 quantitative variables

The analysis begins with the sample variance-covariance matrix of the standardized variables:

$$S = \frac{X^T X}{n-1}$$

which, in our case, coincides with the **symmetric** $p \times p$ correlation matrix, whose diagonal consists only of 1s and correlations elsewhere:

$$R = \frac{Z^T Z}{n-1}$$

	T_{omic}	T_{rapim}	T_{rapine}	$T_{aggress}$	$T_{fcscasso}$	T_{furti}	T_{fauto}
T_{omic}	1	0.60122	0.48371	0.64855	0.38584	0.10192	0.068814
T_{rapim}	0.60122	1	0.59188	0.74026	0.71217	0.61399	0.3489
T_{rapine}	0.48371	0.59188	1	0.55708	0.63726	0.44674	0.59068
$T_{aggress}$	0.64855	0.74026	0.55708	1	0.62292	0.40436	0.27584
$T_{fcscasso}$	0.38584	0.71217	0.63726	0.62292	1	0.79212	0.55791
T_{furti}	0.10192	0.61399	0.44674	0.40436	0.79212	1	0.44418
T_{fauto}	0.068814	0.3489	0.59068	0.27584	0.55791	0.44418	1

TABLE 3. 7x7 Correlation Matrix

The correlation coefficients are quite high, and we know that correlation implies duplication of information. Therefore, it will be possible to use a number of artificial variables smaller than 7.
Since we are working with variables expressed as deviations from the mean, the principal components will not include the initial additive constant, and we can thus write:

$$Y_i = \underline{a_i}^T Z$$

with $\underline{a_i} = (a_{i1}, \dots, a_{ii}, \dots, a_{ip})^T$ and $Z = (Z_1, \dots, Z_p)^T$.

1.2.3. *Construction of the variables.* **Phase I: Identifying** a_1
Y_1 must be determined such that $V(Y_1)$ is maximized under the constraint that $a_1 \cdot a_1^T = 1$. Therefore,

$$\begin{aligned} V(Y_1) &= \frac{\sum_{i=1}^n (Y_{i1} - \overline{Y_1})^2}{n-1} \\ &= \frac{\sum_{i=1}^n (Y_{i1})^2}{n-1} \\ &= \frac{\sum_{i=1}^n (\underline{a_1^T} z_i)^2}{n-1} \\ &= \frac{\sum_{i=1}^n (\underline{a_1^T} z_i)(\underline{a_1^T} z_i)^T}{n-1} \\ &= \frac{\sum_{i=1}^n \underline{a_1^T} z_i z_i^T \underline{a_1}}{n-1} \\ &= \underline{a_1^T} R \underline{a_1} \end{aligned}$$

By imposing the unit norm condition, the function to be maximized will be:

$$\phi(\underline{a_1}, \lambda_1) = \underline{a_1^T} R a_1 - \lambda_1(\underline{a_1^T}\,\underline{a_1} - 1)$$

By taking the derivative, we obtain:

$$\frac{\partial \phi(\underline{a_1}, \lambda_1)}{\partial \underline{a_1}} = 2R\underline{a_1} - 2\lambda_1 \underline{a_1} = 0$$

from which $(R - \lambda_1 I_p)\underline{a_1} = 0$, which is a system of p equations with p unknowns. For it to admit a unique and non-trivial solution, it must hold that $det(R - \lambda_1 I_p) = 0$, which corresponds to finding the eigenvalues and eigenvectors of R.
Let us now consider $R\underline{a_1} = \lambda_1 \underline{a_1}$ and multiply both sides by $\underline{a_1^T}$, obtaining $\underline{a_1^T} R \underline{a_1} = \lambda_1 \underline{a_1^T}\,\underline{a_1}$, which we recognize as $V(Y_1) = \lambda_1$.

Consequently, we search for Y_1 as the largest eigenvalue of R and its corresponding eigenvector.

Phase II: Identifying a_2
Y_2 must be determined such that $V(Y_2)$ is maximized among all combinations uncorrelated with Y_1, under the constraint $\underline{a_2}^T\underline{a_2} = 1$. That is, the following conditions must simultaneously hold:

- $\underline{a_2}^T\underline{a_2} = 1$
- $Cov(\underline{a_1}^T Z, \underline{a_2}^T Z) = 0$

To find $V(Y_2)$, we take the second-largest eigenvalue. Having set up the problem this way, the condition $\underline{a_1}^T\underline{a_2} = 0$ is always satisfied because eigenvectors corresponding to distinct eigenvalues of the same matrix are always orthogonal. This implies that $Cov(Y_1, Y_2) = 0$, since, in general,

$$\begin{aligned} Cov(Y_k, Y_h) &= \frac{\sum_{i=1}^n (Y_{ik} - \overline{Y_k})(Y_{ih} - \overline{Y_h})}{n-1} \\ &= \frac{\sum_{i=1}^n (Y_{ik})(Y_{ih})}{n-1} \\ &= \frac{\sum_{i=1}^n (\underline{a_k}^T Z_i)(\underline{a_h}^T Z_i)}{n-1} \\ &= \frac{\sum_{i=1}^n (\underline{a_k}^T Z_i)(\underline{a_h}^T Z_i)^T}{n-1} \\ &= \underline{a_k}^T [\frac{1}{n-1} \sum_{i=1}^n Z_i Z_i^T] \underline{a_h} \\ &= \underline{a_k}^T R \underline{a_h} \end{aligned}$$

Recalling that $R\underline{a_h} - \lambda_k \underline{a_h} = 0$ and the initial assumption of orthogonality between eigenvectors, we obtain $Cov(Y_k, Y_h) = \lambda_k \underline{a_k}^T \underline{a_h} = 0$.

Phase i: Identifying a_i
Y_i must be determined such that $V(Y_i)$ is maximized among all combinations uncorrelated with $Y_1, \ldots, Y_{i-1}$ under the constraint $\underline{a_i}^T\underline{a_i} = 1$. That is, the following conditions must simultaneously hold:

- $\underline{a_i}^T\underline{a_i} = 1$
- $Cov(\underline{a_i}^T Z, \underline{a_h}^T Z) = 0 \quad \forall h < i$

What can we conclude?
We conclude that the eigenvalues coincide with the variances of the principal components, and specifically:

$$V(Y_1) = \lambda_1 \geq V(Y_2) = \lambda_2 \geq \ldots \geq V(Y_p) = \lambda_p$$

Furthermore, for symmetric matrices like R:

$$tr(R) = \sum_{i=1}^p \lambda_i = \sum_{i=1}^p V(Y_i) = p$$

1.2.4. *SVD: How does it relate?* It is evident that finding the eigenvectors and corresponding eigenvalues of the correlation matrix R is a central problem in PCA. It is known that any symmetric square matrix is diagonalizable. Therefore, since R is symmetric, it can be written as:

$$R = Q\Lambda Q^T$$

where Q is an orthogonal $p \times p$ matrix whose columns are the eigenvectors of R, and Λ is a diagonal $p \times p$ matrix containing the eigenvalues in decreasing order.
SVD helps us retrieve these matrices: it is a matrix factorization technique analogous to diagonalization but applicable to any matrix.
Let $A \in \mathbb{R}^{n\times p}$; then there exist $U \in \mathbb{R}^{n\times n}$ and $V \in \mathbb{R}^{p\times p}$ orthogonal matrices, and $\Sigma \in \mathbb{R}^{n\times p}$ a diagonal matrix such that:

$$A = U\Sigma V^T$$

With some algebraic manipulations, we see that the matrix A^TA, being symmetric $p \times p$, has the SVD decomposition:

$$A^TA = V\Sigma^T\Sigma V^T$$

and the matrix AA^T, being symmetric $n \times n$, has the SVD decomposition:

$$AA^T = U\Sigma\Sigma^T U^T$$

where $\Sigma^T\Sigma$ is a diagonal $n\times n$ matrix containing the values σ_i^2, and $\Sigma\Sigma^T$ is diagonal with non-zero values only in the first k positions, with $k = \text{rank}(A)$.
Returning to $R = \frac{Z^TZ}{n-1}$, it will have the SVD decomposition:

$$\frac{V\Sigma^T\Sigma V^T}{n-1}$$

Thus, the columns of V will be the eigenvectors/coefficients of the linear combinations $\underline{a_i}$, and the eigenvalues will be $\lambda_i = \frac{\sigma_i^2}{n-1}$.
The principal components can then be obtained as:

$$ZV = U\Sigma$$

I will calculate the principal components directly using the SVD of the matrix R, from which I will derive the matrix V and the eigenvalue matrix Λ.

1.2.5. *Application to the dataset.* We calculate the eigenvalues and eigenvectors of the matrix A^TA using the following algorithm:

Algorithm 1 Symmetric QR Method

Require: $A \in \mathbb{R}^{m\times n}$
  Compute A^TA
  $A_0 \leftarrow A$
  $Q \leftarrow I$
  for $k = 0$ to ITMAX: **do**
    Compute $Q_kR_k = A_k$
    $A_{k+1} = R_kQ_k$
    $Q = Q \cdot Q_k$
    Stopping test on the lower triangle of A_k
  end for

By this method, we obtain the diagonal matrix of eigenvalues $A_k = \Lambda$, which corresponds to $\frac{\Sigma^T \Sigma}{n-1}$, the orthogonal matrix of eigenvectors Q, corresponding to the V matrix in the SVD decomposition, and the residual plot as a function of iterations.

λ_1	λ_2	λ_3	λ_4	λ_5	λ_6	λ_7
4.1150	0	0	0	0	0	0
0	1.2387	0	0	0	0	0
0	0	0.7258	0	0	0	0
0	0	0	0.3164	0	0	0
0	0	0	0	0.2580	0	0
0	0	0	0	0	0.2220	0
0	0	0	0	0	0	0.1241

TABLE 4. Eigenvalues of R

a_1	a_2	a_3	a_4	a_5	a_6	a_7
-0.30028	0.62917	-0.17825	0.23215	-0.53807	0.25919	-0.26758
-0.43176	0.16942	0.24419	-0.062293	-0.18859	-0.77327	0.2964
-0.39687	-0.042262	-0.49586	0.55796	0.51998	-0.11454	0.0038374
-0.39665	0.34353	0.06949	-0.62982	0.50665	0.17237	-0.1917
-0.44016	-0.20331	0.20992	0.057688	-0.10092	0.53593	0.64817
-0.35736	-0.40233	0.53922	0.23483	-0.030102	0.039445	-0.60171
-0.29517	-0.50243	-0.56839	-0.41924	-0.36979	-0.057137	-0.14697

TABLE 5. Eigenvectors of R

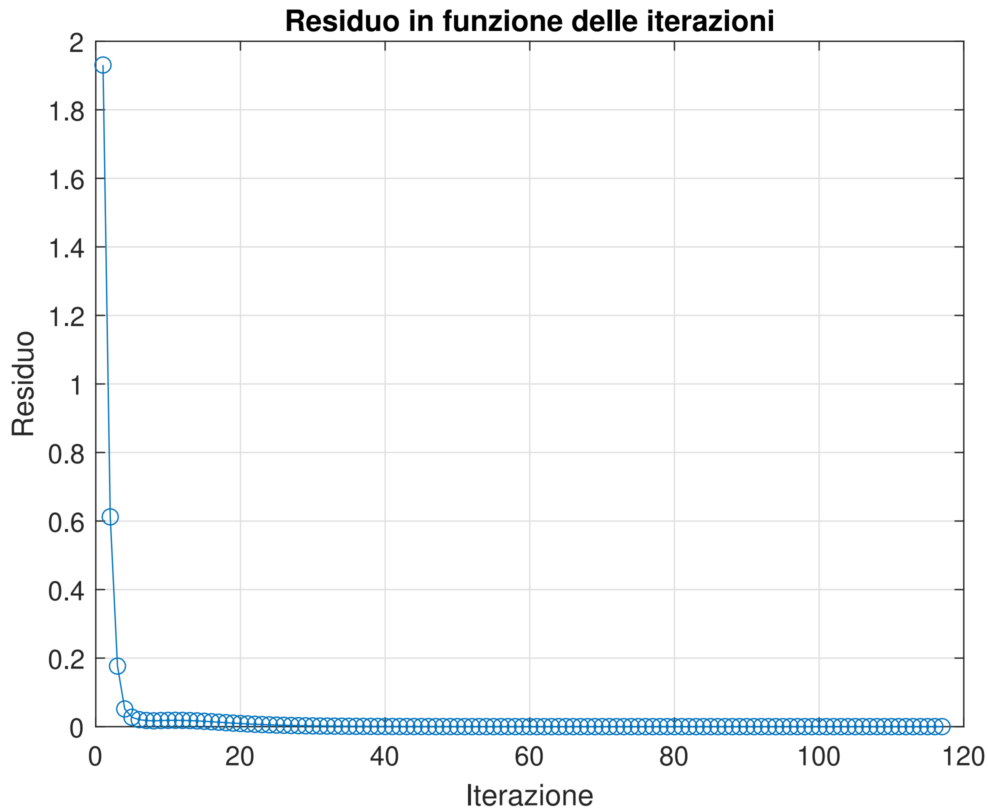

FIGURE 1. We observe that the method has a very high convergence rate

We now select **only** k principal components, corresponding to the number of eigenvalues > 1. In practice, we are approximating the explanation of the variability of the variables of interest. This procedure could have been performed earlier during the SVD computation, obtaining the reduced-rank approximation.
From the matrix (4), we observe that only the first two eigenvalues are > 1, so we limit ourselves to considering the first two principal components:

$$Y_1 = 0.303Z_1 + 0.4318Z_2 + 0.3969Z_3 + 0.3967Z_4 + 0.4402Z_5 + 0.3575Z_6 + 0.2952Z_7$$

$$Y_2 = 0.6292Z_1 + 0.1694Z_2 - 0.0423Z_3 + 0.3435Z_4 - 0.2022Z_5 - 0.4023Z_6 - 0.5024Z_7$$

If $Y_1 > 0$, it indicates that the considered city has an overall danger level above the average, as Y_1 can be considered an indicator of the city's overall danger level.

In Y_2, the coefficients associated with Z_1, Z_2, Z_4 are positive, while those associated with Z_3, Z_5, Z_6, Z_7 are negative. The first three variables relate to crimes such as homicides, kidnappings, and burglaries, which directly compromise individual freedom and typically affect personal safety.
The other four variables relate to crimes such as robberies, break-ins, thefts, and motor vehicle thefts, which are primarily associated with property-related crimes.

If $Y_2 > 0$, the city is predominantly characterized by more violent crimes, while if $Y_2 < 0$, the city is characterized by property-related crimes.

1.2.6. *What did we do statistically?* Simply, we decided to approximate the matrix R with:

$$R_2 = U_{R,(:,1:2)} \cdot \Sigma_{R,(1:2,1:2)} \cdot V^T_{R,(:,1:2)}$$

where $U_{R,(:,1:2)}$ is the matrix of left singular vectors from which we take all rows of the first two columns.

	T_{omic}	T_{rapim}	T_{rapine}	$T_{aggress}$	$T_{fcscasso}$	T_{furti}	T_{fauto}
T_{omic}	0.8614	0.66555	0.45746	0.75785	0.38544	0.12801	-0.02685
T_{rapim}	0.66555	0.80266	0.69626	0.77682	0.73936	0.55048	0.41898
T_{rapine}	0.45746	0.69626	0.65036	0.6298	0.72948	0.60467	0.50835
$T_{aggress}$	0.75785	0.77682	0.6298	0.7936	0.63192	0.41208	0.26798
$T_{fcscasso}$	0.38544	0.73936	0.72948	0.63192	0.84844	0.74859	0.66116
T_{furti}	0.12801	0.55048	0.60467	0.41208	0.74859	0.72601	0.68445
T_{fauto}	-0.02685	0.41898	0.50835	0.26798	0.66116	0.68445	0.67121

TABLE 6. R2

It is demonstrated that the matrix R_2 is, among all rank-2 matrices, the one for which the norm $||R - R_2||$ is minimal and corresponds to the maximum singular value of $R - R_2$. Thus, we derive $||R - R_2|| = 0.7258$. Additionally, I computed this norm for 100,000 random rank-2 matrices, saved the results in a vector, and finally sorted it in ascending order, finding that the minimum of the resulting vector corresponds exactly to 0.7258, in agreement with the previous statement.

1.2.7. *2D Plane Representation of Observations.* We then calculate the scores and loadings. The score of the j-th principal component for the i-th unit is:

$$Y_{ij} = a_{j1}Z_{ij} + \ldots + a_{jp}Z_{ip} \quad \forall i = 1, \ldots, n$$

where a_{js} is the loading of the s-th variable (i.e., the s-th element of the j-th eigenvector).

We then plot the results in the vector space $\mathbb{R}_2$ to visualize the positioning of the statistical units with respect to each possible combination of the two principal components.

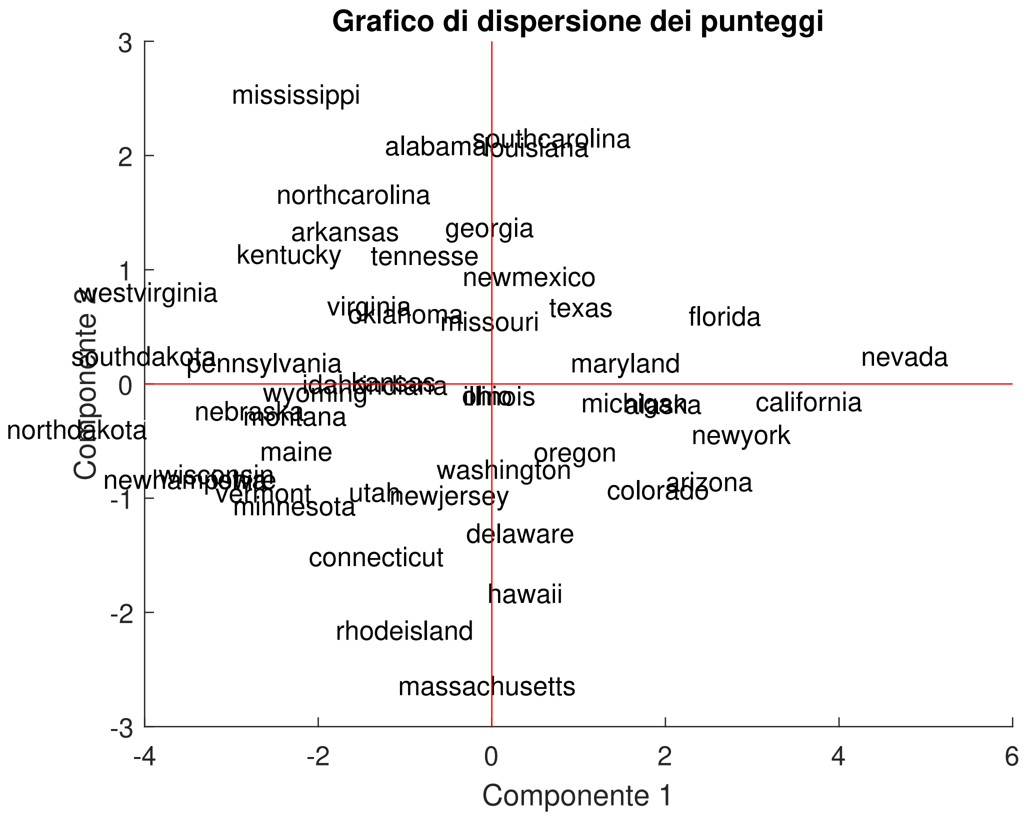

FIGURE 2. Scatter plot of the scores

1.3. **Multivariate Linear Regression Model to Predict the Homicide Rate of a New State.**

1.3.1. *Estimation of the Parameters for the COMPLETE Regression Model.* Given n observations of a set of predictors X and a dependent variable y, a multivariate linear regression model is defined as:

$$y = X\beta + \epsilon$$

where y is an $n \times 1$ vector of the realizations of the dependent variable, $X \in \mathbb{R}^{n\times(p+1)}$ is the data matrix with the first column of ones to account for the intercept, and $\epsilon \in \mathbb{R}^{n\times 1}$ is the random disturbance vector.
It is particularly important to test the following assumptions:

- $E(\epsilon) = 0$
- $\Sigma_\epsilon = \sigma^2 \cdot I_n$

This method aims to explain a linear causal connection between the "predictors" and the response, in addition to enabling simple forecasting.
To estimate the parameters β, the least squares criterion must always be minimized by solving:

$$\phi(\beta) = (y - X\beta)^T(y - X\beta)$$

leading to the system of normal equations:

$$X^T X\beta = X^T y$$

If $(X^T X)^{-1}$ exists, meaning if $rank(X) = p + 1$, then the system has a unique and non-trivial solution. The parameter estimator β is **BLUE** (*best linear unbiased estimator*) and is given by:

$$\beta_{OLS} = (X^T X)^{-1}X^T y$$

Applying the QR method for least squares estimation, we obtain:

$$\hat{\beta}_{OLS} = (0, 0.4445, 0.2690, 0.2806, 0.2511, -0.4958, -0.2424)^T$$

To assess the significance of each predictor, statistical tests such as the *T-Test* or *F-Test* should be applied. To keep the analysis concise, we select the model with the highest $R^2_{adjusted}$, which turns out to be the one including all variables: $R^2_{adjusted} = 0.5544$.

1.3.2. *How does the in-sample error behave?* At this point, we derive the in-sample predictions using our model and compare them with the observed values to evaluate the model's accuracy. A good model should have observations closely aligned along the bisector of the first and third quadrants, and the series of regression residuals should be as random as possible, showing no deterministic pattern.

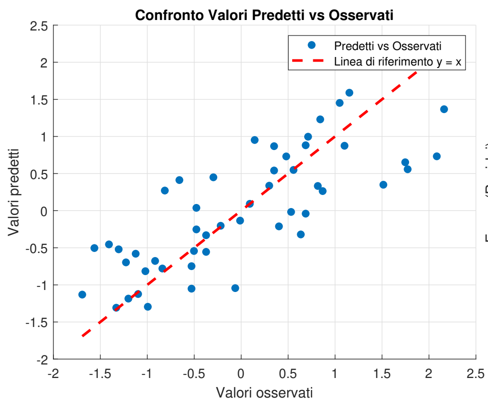

FIGURE 3. Predicted vs Observed Plot

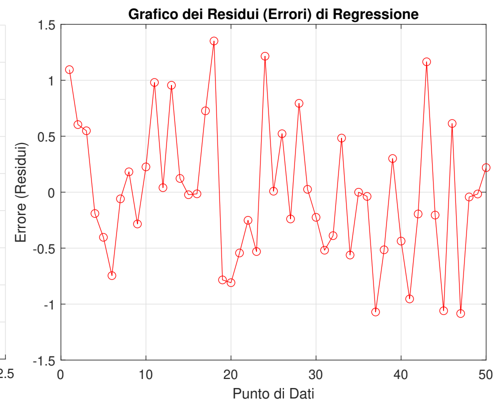

FIGURE 4. Regression Residuals Plot

There seems to be a portion of variance not well explained by the model and still intrinsic in the stochastic component. Indeed, we observe that the process trajectory still appears to have a slight deterministic component, leading us to hypothesize a model misspecification.

1.3.3. *Reduced Regression Model: Using Principal Components*. We now consider the model trained on the principal components Y_1, Y_2, whose OLS parameter estimates are:

$$\beta_{PC_{OLS}} = (0, 0.3003, 0.6292)$$

This model has $R^2_{adjusted} = 0.8555$, which is significantly higher than the model with 7 variables. Thus, we have managed to explain a substantial portion of the variance in the model using only two artificial variables instead of 7.

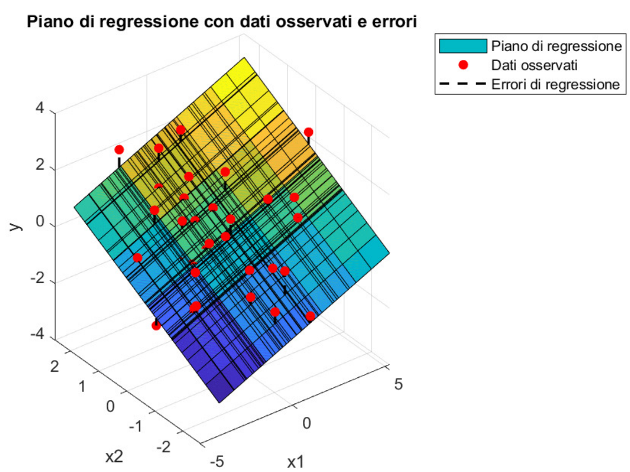

FIGURE 5. Regression Plot in the 2D Plane

1.3.4. *How does the in-sample error behave?* We plot, as before, the behavior of the residuals:

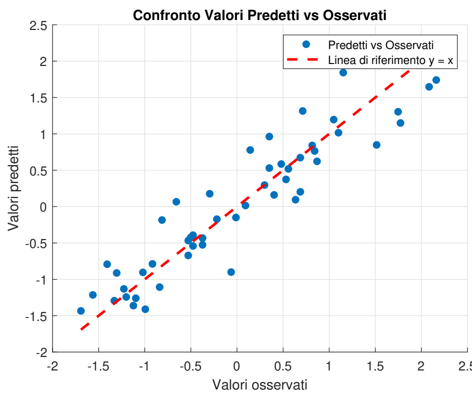

FIGURE 6. Predicted vs Observed Plot

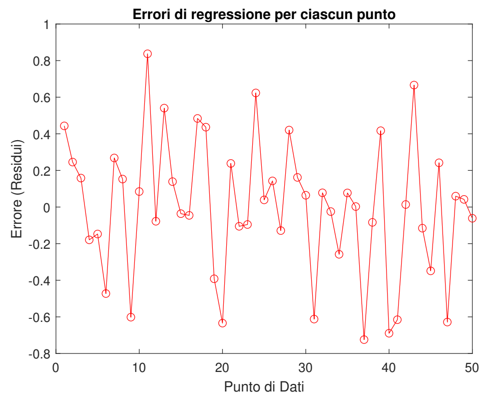

FIGURE 7. Regression Residuals Plot

The residual behaves exactly as desired! It represents the realization of a zero-mean process that is almost entirely random, reflecting the model's goal of capturing all the possible deterministic components!

2. Part II

Imagine having a database containing images of the faces of various criminals, along with a photo of a thief caught in the act by a surveillance camera. The goal is to determine whether the thief's face is already present in the database using a technique known as Face Recognition.

This technique allows us to compare the newly acquired face image with those stored in the database to identify similarities that indicate a match between the thief and one of the recorded faces.

2.1. Model Construction.

Assume that each face in the database has $m \times m = M$ pixels and is represented as a column vector $M \times 1$ called f_i.
The *Training Set* S formed by N images is represented by an $M \times N$ matrix of the form:

$$S = [f_1, \dots, f_N]$$

We now calculate the *mean face* $\overline{f}$ defined as the $M \times 1$ vector:

$$\overline{f} = \frac{\sum_{i=1}^{N} f_i}{N}$$

and center our training data by finding the column vectors:

$$a_i = f_i - \overline{f} \quad \forall i = 1, 2, \dots, N$$

resulting in the centered data matrix of dimension $M \times N$:

$$A = [a_1, \dots, a_N]$$

Let $rank(A) = r \leq N \ll M$. We know that A has the reduced SVD decomposition:

$$A = U \Sigma V^T$$

with $U \in M \times r$ orthogonal, $\Sigma \in r \times r$ diagonal, and $V \in r \times r$ orthogonal. It is then evident that:

$$AV = U\Sigma$$

which implies:

$$Av_i = \sigma_i u_i \quad \forall i = 1, \dots, r$$

It can be shown that the set:

$$u_1, \dots, u_r$$

forms an orthonormal basis for the $range(A)$.

In fact, A can be defined as:

$$A = \sum_{i=1}^{r} \sigma_i u_i v_i^T$$

where each term in the summation is a rank-1 matrix contributing to the construction of A along the direction of u_i.
For any vector y belonging to $range(A)$, there exists a vector x such that:

$$y = Ax = U\Sigma(V^T x)$$

that is:

$$y = \sum_{i=1}^{r} c_i u_i$$

where c_i depends on the i-th component of the vector $V^T x$ and the singular values σ_i.

This shows that any vector in the column space of A can be expressed as a linear combination of the orthonormal set:

$$\{u_1, \ldots, u_r\}$$

which is therefore an orthonormal basis.

We call $range(A)$ the "face subspace" of the space containing all pixel images $m \times n$, referred to as the "image space". Each left singular vector u_i corresponds to a basis vector called a "base-face". It is important to note that these singular vectors do not necessarily correspond to distinct features like ears, eyes, or noses.

Let:

$$x = [x_1, \ldots, x_r]^T$$

be the coordinates of each $m \times n$ face f in the "face subspace". From the preceding discussion, it is evident that these are the scalar projections of $f - \overline{f}$ onto the orthonormal basis vectors, hence:

$$x = [u_1, \ldots, u_r]^T (f - \overline{f})$$

with x being a vector of at most $r \times 1$.

We are thus finding the principal directions of variation in the set of images. This operation is analogous to what PCA does: reducing dimensionality by identifying the main axes of variance. This coordinate vector x is used to find which face in the training set best describes the face f. In other words, it helps identify a training face $f_i \quad \forall i = 1, \ldots, N$ that minimizes the distance:

$$\epsilon_i = ||x - x_i||_2 = [(x - x_i)^T (x - x_i)]^{\frac{1}{2}}$$

where x_i is the coordinate vector of f_i, obtained as the scalar projection:

$$x_i = [u_1, \ldots, u_r]^T (f_i - \overline{f})$$

Problem!!
The objective is to find the left singular values of A, which is $M \times N$, and M is often very large. Instead, we could compute the singular vectors of AA^T, which is $M \times M$, corresponding exactly to the left singular vectors U of A.
However, this computation is also computationally expensive.

The final approach is to calculate the matrix $A^T A$, which has dimensions $N \times N$, and compute its singular vectors corresponding to the right singular vectors of A. We then find V such that:

$$A^T A v_i = \mu_i v_i$$

From the theory of SVD, we know that $A^T A$ and AA^T have the same eigenvalues, and their eigenvectors are related by the formula:

$$AA^T A v_i = \mu_i A v_i$$

from which:

$$A v_i = \mu_i \lambda_i u_i$$

Thus, it suffices to find the eigenvectors of $A^T A$, compute AV, and normalize each column to 1 to obtain the matrix U.

It is important to note that using the comparison between projections is advantageous because, in this way, the algorithm will at most compare the vector x_i of dimension $r \times 1$ with N vectors $r \times 1$, whereas using the original matrices would require comparing an image with M pixels, where $M \gg r$, with N images of M pixels.
The computational gain using truncated SVD is extremely significant.

2.1.1. *Face Recognition on the OLIVETTI Dataset.* The dataset at our disposal is a database of 10 images for 40 different individuals, resulting in $N = 400$. For some subjects, the images were taken at different times, slightly varying the lighting, facial expressions (e.g., open/closed eyes, smiling/not smiling), and facial details (e.g., with/without glasses). All images were taken against a dark, uniform background, and the subjects are in a frontal and vertical position (with some tolerance for slight lateral movement).

The size of each image is 112×92 pixels, with 8-bit grayscale levels. It follows that $m = 112$ and $n = 92$.

FIGURE 8. The 10 photos of individual 1.

With a database split into a *training set* and a *test set* with weights of 0.7 and 0.3 respectively, the following mean face is obtained:

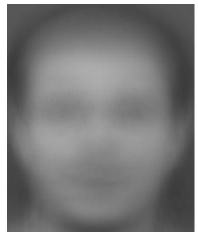

FIGURE 9. Mean face with a training set size of 0.7.

Proceeding as suggested by the theory, the classifier model achieves an accuracy of **98.3333%** using the full-rank matrix.

2.1.2. *Rank-Dependent Accuracies.* It is natural to hypothesize a certain non-linear relationship between the classifier model's accuracy and the regressors $s =$ *training set size* and $r =$ *rank* of the training set.
That is, we expect:

$$\theta = f(s, r)$$

where θ is the model's accuracy.

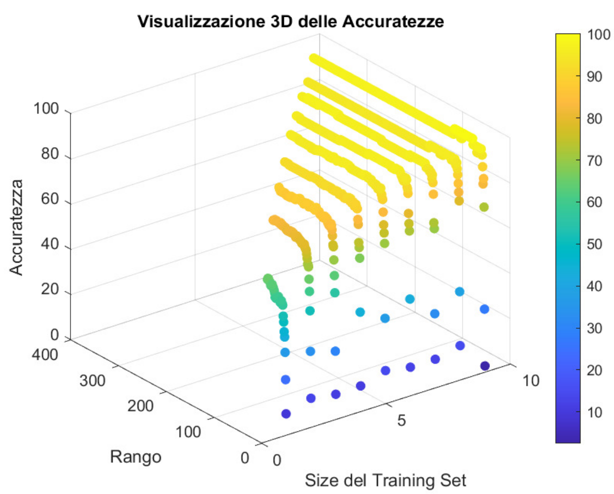

FIGURE 10. Rank and size-dependent accuracies.

Indeed, there exists an exponential relationship that can be formalized as a model of the form:

$$\theta(s, r) = 1 - \theta_0 e^{-g(s,r)}$$

where g is some function of the two variables s and r to be identified.

The function describing the relationship between accuracy θ, rank r, and training set size s can be written as:

$$\theta(s, r) = 1 - k \cdot e^{-\left(\frac{\alpha \cdot s}{\beta + r}\right)}$$

where:

- s represents the training set size,
- r represents the rank,
- k is a scaling parameter for the maximum range of θ,
- α controls sensitivity to the training set size,
- β is a scaling parameter for the effect of rank.

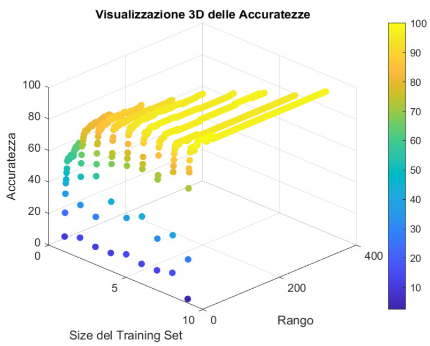

FIGURE 11. Rank and size-dependent accuracies.

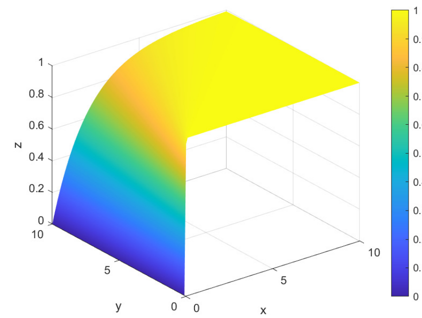

FIGURE 12. Example of a function.

3. Conclusions

The SVD (Singular Value Decomposition) technique has proven to be a highly efficient tool for handling large-scale data matrices, enabling significant reductions in computational costs and memory requirements. However, there are numerous opportunities to enhance and improve the facial recognition algorithm.

One improvement could include a preprocessing phase involving optimal scaling and resizing of input images to ensure greater consistency in face dimensions and features. Additionally, it would be interesting to conduct a systematic comparison between the accuracy of the SVD-based model and that of deep neural network models, such as convolutional neural networks (CNNs) or deep multi-layer networks, evaluating each model's ability to generalize to new data.

An innovative research direction could involve implementing a live facial recognition system via video rather than single snapshots. This could be achieved by combining SVD with real-time face tracking algorithms, analyzing dynamic facial changes, and reducing the incidence of false positives.

In the future, the implementation of facial recognition algorithms based on quantum neural networks (QNNs) could open an even more promising pathway, aiming to achieve unprecedented levels of accuracy and speed. These quantum methods could enable simultaneous analysis of multiple faces in real-time with reduced computational resource consumption.

4. Complete Dataset

	T_{omic}	T_{rapim}	T_{rapine}	$T_{aggress}$	$T_{fcscasso}$	T_{furti}	T_{fauto}
alabama	14.2	25.2	96.8	278.3	1135.5	1881.9	280.7
alaska	10.8	51.6	96.8	284	1331.7	3369.8	753.3
arizona	9.5	34.2	138.2	312.3	2346.1	4467.4	439.5
arkansas	8.8	27.6	83.2	203.4	972.6	1862.1	183.4
california	11.5	49.4	287	358	2139.4	3499.8	663.5
colorado	6.3	42	170.7	292.9	1935.2	3903.2	477.1
connecticut	4.2	16.8	129.5	131.8	1346	2620.7	593.2
delaware	6	24.9	157	194.2	1682.6	3678.4	467
florida	10.2	39.6	187.9	449.1	1859.9	3840.5	351.4
georgia	11.7	31.1	140.5	256.5	1351.1	2170.2	297.9
hawaii	7.2	25.5	128	64.1	1911.5	3920.4	489.4
idaho	5.5	19.4	39.6	172.5	1050.8	2599.6	237.6
illinois	9.9	21.8	211.3	209	1085	2828.5	528.6
indiana	7.4	26.5	123.2	153.5	1086.2	2498.7	377.4
iowa	2.3	10.6	41.2	89.8	812.5	2685.1	219.9
kansas	6.6	22	100.7	180.5	1270.4	2739.3	244.3
kentucky	10.1	19.1	81.1	123.3	872.2	1662.1	245.4
louisiana	15.5	30.9	142.9	335.5	1165.5	2469.9	337.7
maine	2.4	13.5	38.7	170	1253.1	2350.7	246.9
maryland	8	34.8	292.1	358.9	1400	3177.7	428.5
massachusetts	3.1	20.8	169.1	231.6	1532.2	2311.3	1140.1
michigan	9.3	38.9	261.9	274.6	1522.7	3159	545.5
minnesota	2.7	19.5	85.9	85.8	1134.7	2559.3	343.1
mississippi	14.3	19.6	65.7	189.1	915.6	1239.9	144.4
missouri	9.6	28.3	189	233.5	1318.3	2424.2	378.4
montana	5.4	16.7	39.2	156.8	804.9	2773.2	309.2
nebraska	3.9	18.1	64.7	112.7	760	2316.1	249.1
nevada	15.8	49.1	323.1	355	2453.1	4212.6	559.2
newhampshire	3.2	10.7	23.2	76	1041.7	2343.9	293.4
newjersey	5.6	21	180.4	185.1	1435.8	2774.5	511.5
newmexico	8.8	39.1	109.6	343.4	1418.7	3008.6	259.5
newyork	10.7	29.4	472.6	319.1	1728	2782	745.8
northcarolina	10.6	17	61.3	318.3	1154.1	2037.8	192.1
northdakota	0.9	9	13.3	43.8	446.1	1843	144.7
ohio	7.8	27.3	190.5	181.1	1216	2696.8	400.4
oklahoma	8.6	29.2	73.8	205	1288.2	2228.1	326.8
oregon	4.9	39.9	124.1	286.9	1636.4	3506.1	388.9
pennsylvania	5.6	19	130.3	128	877.5	1624.1	333.2
rhodeisland	3.6	10.5	86.5	201	1489	2844.1	791.4
southcarolina	11.9	33	105.9	485.3	1613.6	2342.4	245.1
southdakota	2	13.5	17.9	155.7	570.5	1704.4	147.5
tennessee	10.1	29.7	145.8	203.9	1259.7	1776.5	314
texas	13.3	33.8	152.4	208.2	1603.1	2988.7	397.6
utah	3.5	20.3	68.8	147.3	1171.6	3004.6	334.5
vermont	1.4	15.9	30.8	101.2	1348.2	2201	265.2
virginia	9	23.3	92.1	165.7	986.2	2521.2	226.7
washington	4.3	39.6	106.2	224.8	1605.6	3386.9	360.3
westvirginia	6	13.2	42.2	90.9	597.4	1341.7	163.3
wisconsin	2.8	12.9	52.2	63.7	846.9	2614.2	220.7
wyoming	5.4	21.9	39.7	173.9	811.6	2772.2	282

Table 7. Data matrix: 50 states and 7 quantitative variables

	T_{omic}	T_{rapim}	T_{rapine}	$T_{aggress}$	$T_{fcscasso}$	T_{furti}	T_{fauto}
alabama	1.7472	-0.04963	-0.30891	0.66831	-0.36165	-1.0874	-0.50067
alaska	0.86791	2.404	-0.30891	0.72516	0.092047	0.96226	1.943
arizona	0.53171	0.78683	0.15969	1.0075	2.4377	2.4743	0.32045
arkansas	0.35068	0.17343	-0.46285	-0.078801	-0.73834	-1.1147	-1.0038
california	1.0489	2.1995	1.8439	1.4633	1.9598	1.1413	1.4787
colorado	-0.29585	1.5118	0.52755	0.81394	1.4876	1.6971	0.51488
connecticut	-0.83894	-0.83033	0.061212	-0.79299	0.12511	-0.069689	1.1152
delaware	-0.37344	-0.077512	0.37248	-0.17057	0.90347	1.3874	0.46265
florida	0.71274	1.2887	0.72223	2.372	1.3135	1.6107	-0.13509
georgia	1.1007	0.49872	0.18572	0.45086	0.13691	-0.69029	-0.41173
hawaii	-0.063102	-0.021748	0.044234	-1.4683	1.4328	1.7208	0.57848
idaho	-0.50275	-0.58868	-0.95635	-0.38702	-0.55751	-0.098756	-0.72353
illinois	0.63516	-0.36563	0.98709	-0.022942	-0.47842	0.21657	0.78117
indiana	-0.011379	0.071192	-0.010096	-0.57654	-0.47565	-0.23775	-0.00065152
iowa	-1.3303	-1.4066	-0.93824	-1.2119	-1.1086	0.019027	-0.81505
kansas	-0.21827	-0.34704	-0.26477	-0.30722	-0.049703	0.093692	-0.68888
kentucky	0.68688	-0.61656	-0.48662	-0.87778	-0.9705	-1.3902	-0.68319
louisiana	2.0834	0.48013	0.21288	1.2389	-0.29227	-0.27743	-0.20593
maine	-1.3044	-1.137	-0.96654	-0.41196	-0.089707	-0.44164	-0.67544
maryland	0.14379	0.84259	1.9016	1.4723	0.24998	0.69762	0.26358
massachusetts	-1.1234	-0.45857	0.50944	0.20249	0.55568	-0.49591	3.9431
michigan	0.47999	1.2236	1.5598	0.6314	0.53372	0.67186	0.86856
minnesota	-1.2269	-0.57939	-0.43229	-1.2518	-0.3635	-0.15427	-0.17801
mississippi	1.7731	-0.57009	-0.66093	-0.22144	-0.87014	-1.9719	-1.2054
missouri	0.55757	0.23848	0.73468	0.22144	0.061061	-0.34038	0.0045193
montana	-0.52861	-0.83962	-0.96088	-0.54362	-1.1261	0.14039	-0.3533
nebraska	-0.91653	-0.7095	-0.67225	-0.98351	-1.23	-0.4893	-0.66406
nevada	2.161	2.1716	2.2525	1.4334	2.6852	2.1233	0.9394
newhampshire	-1.0976	-1.3973	-1.142	-1.3496	-0.57855	-0.451	-0.435
newjersey	-0.47688	-0.43998	0.63734	-0.26134	0.33277	0.14218	0.69275
newmexico	0.35068	1.2422	-0.16403	1.3177	0.29323	0.46468	-0.61029
newyork	0.84205	0.34072	3.9447	1.0753	1.0085	0.15252	1.9043
northcarolina	0.81619	-0.81174	-0.71073	1.0673	-0.31863	-0.87268	-0.9588
northdakota	-1.6924	-1.5553	-1.254	-1.6708	-1.9558	-1.141	-1.2039
ohio	0.092067	0.14554	0.75166	-0.30124	-0.1755	0.035145	0.11828
oklahoma	0.29896	0.32213	-0.56925	-0.062841	-0.008542	-0.61053	-0.26229
oregon	-0.65791	1.3166	9.055e-05	0.75409	0.79664	1.15	0.058812
pennsylvania	-0.47688	-0.62586	0.070267	-0.8309	-0.95824	-1.4426	-0.2292
rhodeisland	-0.99411	-1.4158	-0.4255	-0.10274	0.45579	0.23806	2.1401
southcarolina	1.1524	0.6753	-0.20591	2.7331	0.74391	-0.45307	-0.68475
southdakota	-1.4079	-1.137	-1.202	-0.5546	-1.6682	-1.332	-1.1894
tennessee	0.68688	0.3686	0.24571	-0.073813	-0.074445	-1.2326	-0.32848
texas	1.5144	0.74965	0.32041	-0.030922	0.71963	0.43726	0.1038
utah	-1.02	-0.50504	-0.62584	-0.63838	-0.27817	0.45917	-0.22248
vermont	-1.5631	-0.91397	-1.056	-1.0982	0.1302	-0.64786	-0.58081
virginia	0.4024	-0.22622	-0.36211	-0.45485	-0.70689	-0.20676	-0.77989
washington	-0.81308	1.2887	-0.20252	0.13466	0.72541	0.98582	-0.089072
westvirginia	-0.37344	-1.1649	-0.92692	-1.201	-1.6059	-1.8316	-1.1077
wisconsin	-1.201	-1.1928	-0.81373	-1.4723	-1.029	-0.078643	-0.81091
wyoming	-0.52861	-0.35633	-0.95522	-0.37306	-1.1106	0.13901	-0.49394

TABLE 8. Standardized data matrix: 50 states and 7 quantitative variables